SCSQ-Net: A Shared Kernel Point Convolution Semantic Query Network for Weakly Supervised Classification of Multispectral LiDAR Point Clouds

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***Abstract*— Multispectral LiDAR (MS-LiDAR) point cloud classification holds great potential, but current methods rely heavily on fully supervised learning, requiring costly manual labeling. To address this, we propose a shared convolutional semantic query network (SCSQ-Net), a novel weakly supervised model designed for airborne MS-LiDAR data. It utilizes a shared kernel point convolution (SKPC) backbone to integrate neighborhood information during both encoding and semantic inference. An adaptive dropout activation (ADA) module is introduced to further enhance activation and feature learning across the entire point cloud scene. Our loss strategy combines point affinity loss, weighted cross-entropy loss, and Lovasz–Softmax loss for comprehensive supervision. Evaluated on two large datasets, the SCSQ-Net demonstrated performance comparable to fully supervised models, achieving an average $F1$-score of 86.83%, an intersection over union (mIoU) of 80.71%, and overall accuracy (OA) of 96.29% using only 0.1% labeled data.**

***Index Terms*— Dropout activation, kernel point convolution, multispectral LiDAR (MS-LiDAR), point cloud classification, weakly supervised network.**

I. Introduction

MULTISPECTRAL LiDAR (MS-LiDAR) data, as a data source of active remote sensing, offer both precise 3-D representations and rich spectral information of large-scale outdoor scenes, which has significant potential for applications in various fields, including geological surveying [1], coastal management [2], [3], forest monitoring [4], [5], and land cover mapping [6], [7], [8]. These applications rely on point cloud classification, involving the extraction of semantic information from MS-LiDAR data.

In recent years, deep learning techniques applied to MS-LiDAR data have made significant progress. Inspired by methods of processing conventional single-channel LiDAR data [9], [10], [11], [12], [13], novel approaches that achieved state-of-the-art (SOTA) results have consistently emerged. SE-PointNet++ [14] integrates a squeeze-and-excitation block (SE-block) into PointNet++ [10]. It improves the classification performance of MS-LiDAR point clouds by modeling the interdependencies among channels. FR-GCNet [15] first utilizes a farthest point sampling (FPS) and K-nearest neighbors (KNN) (FPS-KNN) sampling to generate training samples, effectively preserving geometric structures of objects in an MS-LiDAR point cloud. Next, FR-GCNet learns representative local features from the dependencies between points in the neighborhood graph relationships and infers global contextual relationships from an MS-LiDAR point cloud, achieving efficient classification performance. Li et al. [16] incorporated geometric moment features into a graph convolutional network (GCN) to enhance point cloud feature descriptions, accurately extracting buildings from MS-LiDAR point clouds. Subsequently, the team proposed an attention-based geometric moment convolution operator [17], which integrates multiscale MS-LiDAR point cloud features in a feature pyramid fusion way, further improving the network to describe details at various scales. Xiao et al. [18] integrated local geometries and feature content into the RandLA-Net [13] based on 3-D spatial geometric associations, achieving satisfactory MS-LiDAR classification performance. MS-AMCNN [19] utilizes a local adjacency matrix convolution module designed to fully exploit the topological relationships of point clouds while perceiving local geometric features. Although these approaches have achieved promising results in the classification of MS-LiDAR point clouds, most of them heavily rely on a large volume of precisely annotated data. Unfortunately, annotating 3-D point clouds in large and complex scenes is time-consuming and labor-intensive. The multichannel spectral information of MS-LiDAR data further intensifies this challenge and increases

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the visual interpretation complexity. Therefore, data annotation has become a bottleneck hindering the advancement of point cloud classification methods [20].

With the advancements in MS-LiDAR technology and the diversification of data-collection platforms, such as airborne, vehicle-mounted, and handheld systems, the acquisition of point clouds has become significantly easier, enabling the capture of millions of dense points within a short time. If it were possible to achieve satisfactory performance by annotating only a small fraction of such large volume and dense points, this approach would greatly reduce the cost of labeling and promote the application of MS-LiDAR data. In fact, experiments by SQN [21] have demonstrated that even with a 0.1% sparse labeling rate, the performance of the models only experiences a slight decline compared with fully annotated single-channel point clouds. Similarly, FPWS-Net [22] has proven that a weakly supervised method on MS-LiDAR data exhibits comparable effectiveness. There are few weak supervision methods for MS-LiDAR data, but numerous scholars have proposed a range of weakly supervised methods to address the complex annotation challenges when processing single-channel point clouds. These methods are typically categorized into two distinct groups: 1) the use of additional information and 2) the creation of extra-supervised signals.

The former refers to the generation or prediction of auxiliary information that is more readily accessible yet relevant to point cloud classification tasks [20], [23], [24], [25]. This strategy aims to augment model training. Xu and Lee [20] performed point cloud classification with ten times fewer labels by predicting auxiliary augmented samples along with utilizing spatial and color smoothness constraints, thereby providing additional constraints for the network. Furthermore, PSD [23] utilizes an attribute attention module to generate auxiliary augmented samples and constrains these augmented samples with the original samples through Jensen–Shannon divergence constraints, thereby enhancing the supervised signal. It then integrates a GCN into the decoder to model the feature relationships, thereby providing more information for the network. HybridCR [26] aims to enhance the diversity of auxiliary augmented samples using a dynamic point augmentor and combines local and global guided contrastive losses to provide additional constraints for the model. Although these methods can effectively improve the classification performance of point clouds, the use of a large amount of additional input information increases memory consumption.

The latter refers to automatically extracting supervised signals from existing data [27], [28], [29], [30], [31], [32], [33]. Although these signals are not as precise as manually annotated labels, they still provide effective training support without increasing the cost of annotation. For instance, a pretrained classification network assigns pseudolabels to subclouds, utilizing the overlapping regions of subclouds to supervise point cloud classification [27], [28]. Wang and Yao [29] obtained potential information from unlabeled data through entropy regularization to reduce class overlap in prediction probabilities. They also introduced a consistency constraint to enhance the robustness of predictions, and an online soft pseudolabeling strategy to efficiently create additional supervisory sources in a parameter-free manner. Inspired by Wang and Yao [29], WSPointNet [30] with a three-branch structure and adaptive constraints was proposed. It provides the network with efficient adaptive supervised signals, thereby improving the performance of weakly supervised networks. To achieve point cloud classification with minimal labeling, OCOC [31] was proposed as an active weakly supervised framework, including a context-aware pseudolabeling strategy and uncertainty measures to enable low-cost annotation. However, these methods rely on the predictive accuracy of their backbone, and the pseudolabel noise is considerable to greatly mislead network training.

To reduce the computational burden caused by additional samples and supervise the model with a few accurately supervised signals, we, inspired by Hu et al. [21] that leveraged semantic similarity to provide more underlying supervised signals, propose a shared convolutional semantic query network (SCSQ-Net), for classifying airborne MS-LiDAR point clouds. First, to better integrate the neighborhood information of labeled points, the SCSQ-Net shares kernel point convolutions between the encoding path and the semantic inference path. Next, an enhanced feature (EF) module is embedded into the encoding path to supplement local contextual relationships. Furthermore, we develop an adaptive dropout activation (ADA) module, including a spatial dropout block and an activation block. The ADA module adversarially activates the entire point cloud scene to prevent the network from paying excessive attention to discriminative regions. Finally, we implement a loss strategy that combines point affinity loss, weighted cross-entropy loss, and Lovasz–Softmax loss [34], comprehensively supervising network training with only a few labeled points.

The main contributions of this article are as follows.

1) We propose a novel network architecture that innovatively shares kernel point positions and weights between the encoding path and the semantic inference path, providing accurately integrated features for semantic inference.
2) We develop a new ADA module that adaptively suppresses confusing points and activates the remaining areas, enabling the network to focus on the entire point cloud scene.
3) Experiments on two large MS-LiDAR point cloud datasets demonstrate that the proposed method achieved performance comparable to fully supervised models with only 0.1% labeled data.

II. Methodology

In this section, we first present the architecture of the proposed SCSQ-Net, followed by the proposed key modules, i.e., EF module, shared kernel point convolution (SKPC) module, and ADA module. Next, the adopted loss functions employed during the whole training process are presented.

A. SCSQ-Net Architecture

Fig. 1 shows the architecture of the SCSQ-Net. The proposed SCSQ-Net shares kernel point convolutions between

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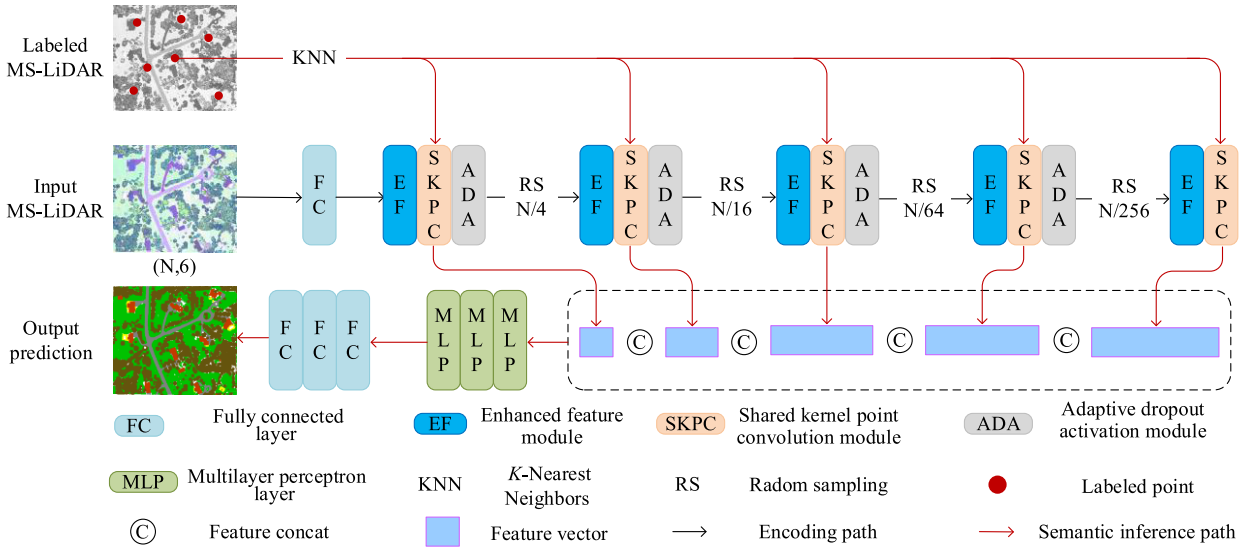


Fig. 1. Illustration of the proposed SCSQ-Net.

an encoding path and a semantic inference path, facilitating the effective integration of high-dimensional features from neighboring and labeled points. Generally, in the encoding path, the SKPC module aggregates local features, with kernel point positions and weights continuously evolving during network training. The semantic inference path directly utilizes the SKPC module from the encoding path, i.e., it integrates information from the labeled point neighborhoods using the same kernel positions and weights as the encoding path.

To accommodate MS-LiDAR point clouds with varying numbers of channels, we first construct a fully connected (FC) layer to uniformly expand the feature dimension of the input to 8. Subsequently, the output features of the FC are input to the encoding path to extract high-dimensional feature information and reduce the point cloud density. The encoding path contains five stages. Each of the first four stages includes an EF module, an SKPC module, and an ADA module, while the last stage includes the EF and SKPC modules. During the five stages, four random samplings are required to reduce the point cloud density and expand the receptive field. Specifically, the EF module aims to model the geometric distance relationship and feature distance relationship between input points and their neighbors. The SKPC module effectively fuses the geometric and spectral information of point clouds, aggregating local information and providing rich and accurate supervised signals for weakly supervised networks. The ADA module is embedded in the encoding path to adaptively dropout discriminative regions and adversarially activate the whole scene, achieving robust point cloud classification performance. The five SKPC modules along the encoding path aggregate local information of point clouds, gradually increasing the feature dimensions ($8 \rightarrow 16 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512$). The sampling rates of the four random samplings are set to 1/4, 1/16, 1/64, and 1/256, respectively.

During the semantic inference path, we leverage the semantic similarity of local points in MS-LiDAR point clouds. At the high-dimensional layer of each encoding stage, nearest neighbor queries (i.e., KNN) are used to find the neighbors for each labeled point. For each labeled point, the SKPC module is used again to aggregate the high-dimensional features of its neighbors in Euclidean space. For each kernel point, the positions and weights used in the SKPC module are consistent with those in the corresponding encoding stage. Subsequently, the aggregated features from the five stages are concatenated together to form the unique feature representation of the labeled point. Following this, three multilayer perceptrons (MLPs) gradually reduce the feature dimensions ($256 \rightarrow 128 \rightarrow 64$), ultimately outputting the feature information of the labeled points. Finally, we construct three FC layers to output the prediction results of the models. Notably, during training, the network infers only the classes of labeled points; however, during testing, we infer the classes of all the point clouds.

Finally, to achieve comprehensive supervision of the sparse labeled points for the proposed SCSQ-Net training, we integrate an affinity loss, a weighted cross-entropy loss, and a Lovasz–Softmax loss.

B. EF Module

Extracting accurate and rich point cloud features is crucial for weakly supervised networks. Traditional kernel point convolution directly assigns the feature information of neighboring points to the kernel points, easily losing important geometric and feature relationships between a central point and its neighboring points. To address this issue, an EF module is designed to supplement the SKPC module.

For each input point $p_i \in \mathbf{R}^{N\times 3}$ with feature f_i, where N denotes the number of input points, we use the KNN to obtain its K_1 neighboring points p_i^{k1} with features f_i^{k1}, where $k1 = \{1, 2, \ldots, K_1\}$. Fig. 2 shows the illustration of the EF module. The input data of the EF module include p_i and its feature f_i, along with the neighbors p_i^{k1} and their features f_i^{k1}.

First, the EF module calculates the geometric distance $p_i^{k1} - p_i$ between p_i and its neighboring points p_i^{k1}, concatenates the

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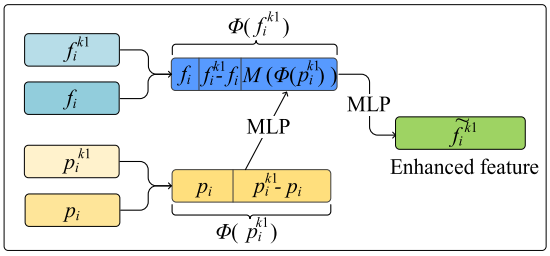


Fig. 2. Diagram of the EF module.

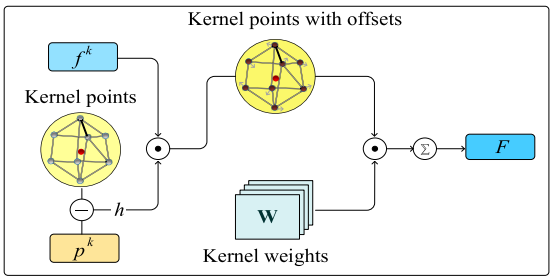


Fig. 3. Diagram of the SKPC module.

geometric distance and input point p_i, denoted as $\phi(p_i^{k1})$, and then inputs $\phi(p_i^{k1})$ into an MLP to output features $M(\phi(p_i^{k1}))$.

Then, the EF module calculates the feature distance $f_i^{k1} - f_i$ between feature f_i of p_i and features f_i^{k1} of p_i^{k1}, and concatenates them with $M(\phi(p_i^{k1}))$ to generate local feature information $\phi(f_i^{k1})$.

Finally, an MLP is used to reduce the dimension of $\phi(f_i^{k1})$ to the same as the original feature dimensions of f_i^{k1}, and then outputs the EFs $\widetilde{f_i^{k1}}$ of neighboring points p_i^{k1}.

These EFs not only include their inherent features but also incorporate local positional and feature relationships of the neighborhood, thereby providing rich and strong semantic information for point cloud classification. The EFs $\widetilde{f_i^{k1}}$ can be formulated as

$$\widetilde{f_i^{k1}} = M\left(\text{concat}\left(M\left(\phi\left(p_i^{k1}\right)\right), f_i^{k1} - f_i, f_i\right)\right) \tag{1}$$

where M represents MLP operation, and concat represents array concatenation.

C. SKPC Module

Fig. 3 details the SKPC module's architecture. As illustrated in Fig. 3, the SKPC module first defines a set of J kernel points x_j, $j = \{1, 2, \ldots, J\}$ within a unit sphere in Euclidean space. This module then employs a linear correlation function to compute the spatial relationships between points p^k ($k = \{1, 2, \ldots, K\}$) and kernel points x_j. The linear correlation function h is written by

$$h\left(p^k, x_j + \Delta_j(x)\right) = \max\left(0, 1 - \frac{||p^k - \left(x_j + \Delta_j(x)\right)||}{d}\right) \tag{2}$$

where $h()$ represents the linear correlation function. $\Delta_j(x)$ represents the learnable offsets of the kernel points that can be adjusted during training. d denotes the important radius of the kernel points. $||\cdot||$ denotes taking the Euclidean distance.

Treating the calculated spatial relationship as weights, high-dimensional features f^k of points p^k are assigned to the corresponding kernel points. Then, the features of the kernel points are multiplied by a kernel weight matrix learned from training and summed over the feature dimensions. Finally, the resulting convolved point cloud features serve as the aggregated local features F. The aggregated local feature F is written by

$$F = \sum_{k=1}^{K} \sum_{j=1}^{J} \mathbf{W} h\left(p^k, x_j + \Delta_j(x)\right) f^k \tag{3}$$

where $\mathbf{W}$ is the kernel weight matrix.

When the SKPC modules are used in the encoding path, the inputs to the SKPC module, p^k and f^k, are the neighboring points p_i^{k1} of p_i and their EFs $\widetilde{f_i^{k1}}$, respectively. When the SKPC modules are used in the semantic inference path, the inputs p^k and f^k are the neighboring points p_l^{k2} of the labeled point $p_l \in \mathbf{R}^{M\times 3}$ and their features f_l^{k2}, where $k2 = \{1, 2, \ldots, K_2\}$. Here, K_2 represents the number of query neighboring points for each labeled point using KNN, and M represents the number of labeled points, significantly less than the number of input points, N.

Although each stage in the encoding path uses the same SKPC module with the corresponding stage in the semantic inference path, the kernel points are redefined at different stages. Specifically, along the encoding path, the SKPC module is applied to each input point, with EFs assigned to the kernel points. In contrast, along the semantic inference path, the module is used exclusively on labeled points. Additionally, to manage memory consumption, the features of the neighboring points of a labeled point are directly assigned to the kernel points.

D. ADA Module

Traditional weakly supervised networks for point clouds often focus excessively on distinctive points rather than the entire point cloud scene. This tendency leads to inaccurate classification results when encountering points with similar features but from different classes. To address this issue, we propose an ADA module, which encourages weakly supervised networks to activate the entire point cloud scene instead of relying solely on discriminative points. Fig. 4 presents the diagram of the ADA module. As illustrated in Fig. 4, the ADA module uses a spatial dropout block to adaptively suppress discriminative regions, followed by an activation block employed to activate the remaining relevant regions. This module allows for extracting features from the entire point cloud, facilitating the training of a robust network.

Concretely, given an input point cloud $p_i \in \mathbf{R}^{N\times C}$, where N denotes the number of points input into the spatial dropout block, and C represents the number of feature channels. For each point cloud $p_i (i = 1, 2, \ldots, N)$, the feature intensity is first normalized, treated as an importance score $\widetilde{s_i}$. The

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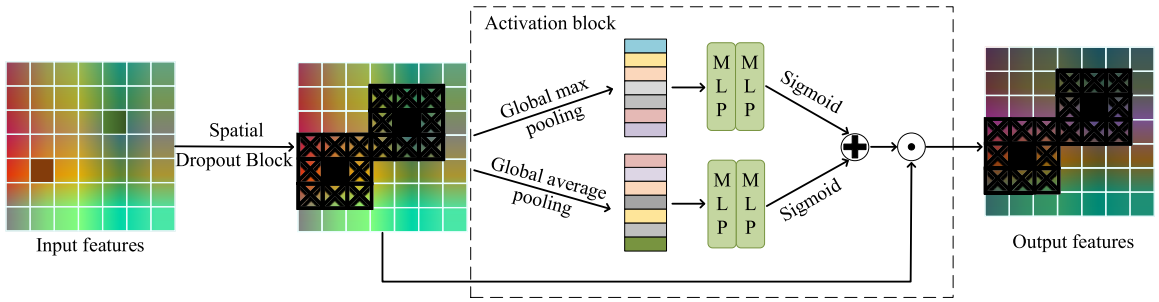


Fig. 4. Diagram of the ADA module.

importance score $\widetilde{s_i}$ is written by

$$s_i = \frac{1}{C}\sum_{j}^{C} |p_{ij}| \tag{4}$$

$$\widetilde{s_i} = \frac{s_i - \min s_k}{\max s_k - \min s_k} \tag{5}$$

where p_{ij} is the value of the jth feature of p_i, and s_i is the feature score of p_i. $\min s_k$ and $\max s_k$ denote the minimum and maximum values among all feature scores, respectively. This normalization method scales all scores to a range between 0 and 1, enhancing the comparability of the scores. Importance score $\widetilde{s_i}$ determines whether the corresponding point cloud region will be discarded. The discard function, denoted as $D(p_i)$, is defined as follows:

$$D(p_i) = \begin{cases} 0, & \widetilde{s_i} > \varphi + 2\sigma \\ 1, & \widetilde{s_i} \le \varphi + 2\sigma \end{cases} \tag{6}$$

$$\varphi = \frac{1}{N}\sum_{i=1}^{N} \widetilde{s_i} \tag{7}$$

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\widetilde{s_i} - \frac{1}{N}\sum_{i=1}^{N}\widetilde{s_i}\right)^2} \tag{8}$$

where φ and σ represent the mean and standard deviation of the importance scores, respectively. If the importance score of p_i exceeds $\varphi + 2\sigma$, $D(p_i) = 0$, otherwise, $D(p_i) = 1$. If $D(p_i) = 0$, we center a square spatial mask with a side length of 3 at p_i, setting the feature values of all points within this region to 0. This method, by removing point features with high importance scores, forces the model to learn a broad range of data characteristics, thereby enhancing its generalization ability to some extent.

To mitigate the negative impact of removing important discriminative regions on model performance, we introduce a feature attention module as the activation block to adaptively enhance the features of the remaining point cloud regions. This design not only activates the overall point cloud features under an adversarial paradigm but also establishes a complementary mechanism within the model. The specific operations are as follows. First, point cloud features processed through the spatial dropout block are input to both global average pooling and global max pooling to obtain two sets of pooled point cloud features. Subsequently, the two pooled features are processed through two shared MLPs for dimensionality reduction and expansion, which effectively achieves feature attention and significantly reduces model complexity. Finally, the processed point cloud features are transformed into weights for each feature through a sigmoid function. These weights are further combined with the input point cloud features to generate the final activated point cloud features.

The coordination between the spatial dropout block and the activation block forms an adversarial mechanism, where the dropout block aims to suppress the distinctive regions, while the activation model strives to emphasize the remaining critical regions. The ADA module effectively mines high-quality point clouds for training a robust weakly supervised network.

E. Loss Functions

Inspired by Zhang et al. [23], we introduce a point affinity loss to fully leverage a few supervised signals and better handle sparsely labeled points. Through affinity loss, our model minimizes the feature differences of labeled points belonging to the same class and enhances predictive capabilities by learning the similarities from the labeled samples via an affinity matrix.

Specifically, given one labeled point $p_l^i \in \mathbf{R}^M$ with a high-dimensional feature vector e_l^i, and another labeled point $p_l^j \in \mathbf{R}^M$ with a high-dimensional feature vector e_l^j, where M represents the number of labeled points, the similarity a_l^{ij} between the labeled points p_l^i and p_l^j is calculated by a cosine similarity. Then, we employ a point correlation descriptor matrix, $\mathbf{C_g}$, to supervise the affinity of labeled points. The functions can be written by

$$a_l^{ij} = \frac{\left\langle e_l^i, e_l^j \right\rangle}{\left|\left|e_l^i\right|\right|_2, \left|\left|e_l^j\right|\right|_2} \tag{9}$$

$$\mathbf{C_g} = \mathbf{V}\mathbf{V}^{\mathbf{T}} \tag{10}$$

where $||\cdot||_2$ denotes l_2 normalization, $\langle\cdot\rangle$ denotes the inner product, $\mathbf{V}$ represents the label matrix after one-hot encoding, and $\mathbf{V^T}$ is the transpose of $\mathbf{V}$. An element c_{ij} of matrix $\mathbf{C_g}$ denotes the semantic relationship between labeled points p_l^i and p_l^j.

Similar to [23], we employ a composite loss L_{af} composed of a cross-entropy loss L_{ce}, a precision loss L_p, and a recall loss L_r to guide the learning of point affinity. The composite loss function allows for a balanced approach to training

$$L_{\text{af}} = L_{\text{ce}} + L_p + L_r \tag{11}$$

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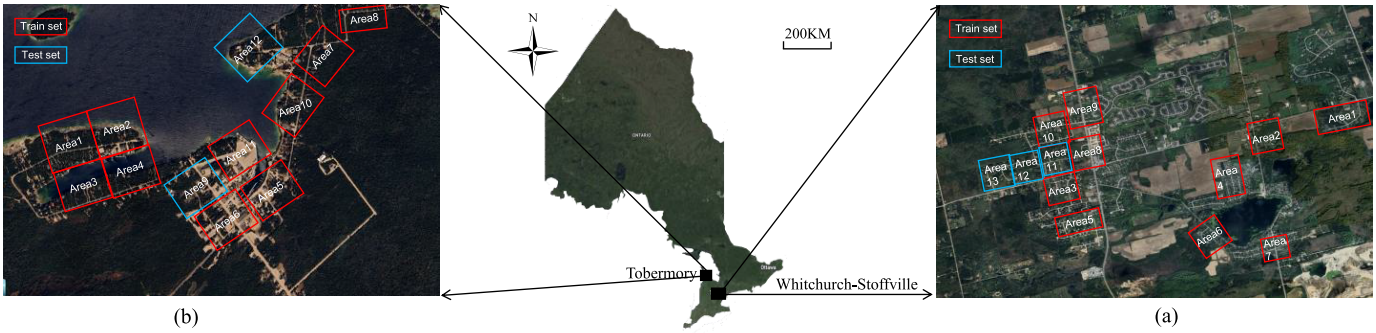


Fig. 5. Illustration of two study areas located in Ontario, Canada. (a) Whitchurch–Stoffville, with central coordinates W79°15′00″, N43°58′00″. The study area (b) is Tobermory, with a central coordinate of W81°39′57″, N45°15′24″.

$$L_{\text{ce}} = -\frac{1}{M^2}\sum_{i=1}^{M}\sum_{j=1}^{M}\left(\mathrm{c}_{ij}\log a_l^{ij} + \left(1-\mathrm{c}_{ij}\right)\log\left(1-a_l^{ij}\right)\right) \tag{12}$$

$$L_p = -\frac{1}{M}\sum_{j=1}^{M}\log\frac{\sum_{i=1}^{M}\mathrm{c}_{ij}a_l^{ij}}{\sum_{i=1}^{M}a_l^{ij}} \tag{13}$$

$$L_r = -\frac{1}{M}\sum_{j=1}^{M}\log\frac{\sum_{i=1}^{M}\mathrm{c}_{ij}a_l^{ij}}{\sum_{i=1}^{M}\mathrm{c}_{ij}}. \tag{14}$$

The final loss function L_{total} includes a weight cross-entropy loss L_{wc}, a Lovasz–Softmax loss function L_{lz}, and an affinity loss L_{af}, and is written by

$$L_{\text{total}} = L_{\text{wc}} + L_{\text{lz}} + L_{\text{af}} \tag{15}$$

$$L_{\text{wc}} = -\frac{1}{M}\sum_{i=1}^{M}w_i\sum_{g=1}^{G}y_{ig}\log\frac{\exp\left(p_l^{ig}\right)}{\sum_{g=1}^{G}\exp\left(p_l^{ig}\right)} \tag{16}$$

$$L_{\text{lz}} = \frac{1}{G}\sum_{g=1}^{G}\overline{\Delta J_g}\left(M_g\right) \tag{17}$$

where w_i denotes the class weight of labeled point i, and is an inverse proportion of the class i to the total points. G denotes the number of classes. y_{ig} denotes the ground-truth labels and p_l^{ig} denotes the probability that the labeled point i belongs to class g. $\overline{\Delta J_g}(M_g)$ denotes the Jaccard coefficient constructed by the error distribution probability vector M_g of samples belonging to class g.

III. Experimental Setting

To evaluate the performance of the SCSQ-Net, we conducted a set of experiments on two datasets, i.e., Whitchurch–Stouffville (WS) and Tobermory MS-LiDAR datasets. Additionally, we compared the SCSQ-Net with nine models, including two fully supervised networks and one weakly supervised network designed for MS-LiDAR data, as well as four fully supervised networks and two weakly supervised networks for conventional single-channel point clouds.

A. Datasets Description

Both of these two datasets were collected in 2015 using an Optech Titan airborne MS-LiDAR system. The standard configuration of the Optech Titan MS-LiDAR system includes three independent laser channels, i.e., a mid-infrared channel (1550 nm), a near-infrared channel (1064 nm), and a green channel (532 nm). The three laser channels are arranged at 3.5° angular intervals, with each channel having an average point density of approximately 3.6 pts/m^2.

The study area of the WS dataset is located in the town of WS, Ontario, Canada, with the central geographic coordinates of longitude 79°15′00″ and latitude 43°58′00″. From the WS MS-LiDAR flight strips, we extracted 13 representative scenes, as shown in Fig. 5(a). Areas 1–10 were selected as training scenes, and Areas 11–13 as test scenes. There are six different land cover and land use types, i.e., road, building, grass, tree, soil, and powerline, which were manually labeled according to high-resolution satellite images of the surveyed study area.

The study area of the Tobermory dataset is located at the northern tip of the Bruce Peninsula, Ontario, Canada, with the central geographic coordinates of longitude 81°39′57″ and latitude 45°15′24″. We selected 12 representative scenes from the collected Tobermory MS-LiDAR flight strips as the Tobermory dataset, as shown in Fig. 5(b). Areas 9 and 12 were selected as testing data, while the others were used as training data. The Tobermory dataset was categorized into six land cover classes, i.e., road, building, land (soil and grass), tree, water, and powerline.

1) Data Fusion and Labeling: As shown in Fig. 6, we used 3-D spatial registration techniques [35] to fuse the individual point clouds collected by the three channels into a single-point cloud containing spectral information from all three channels. We then performed point-by-point manual labeling for the two datasets. Finally, each point in all point cloud scenes contains the following seven attributes: 3-D coordinates (x, y, z), intensity information from the three channels (I_{c1}, I_{c2}, I_{c3}), and ground-truth labels (Label).

2) Sample Generation: Following the sample generation and sparse labeling strategy proposed by [13], we sampled the two datasets with a grid sampling size of 0.5 m. From the sampled data, we randomly selected a fixed proportion (i.e., 0.05% or 0.1%) of total points to serve as sample points and preserve their true labels. The labels of the remaining points were assigned to 0.

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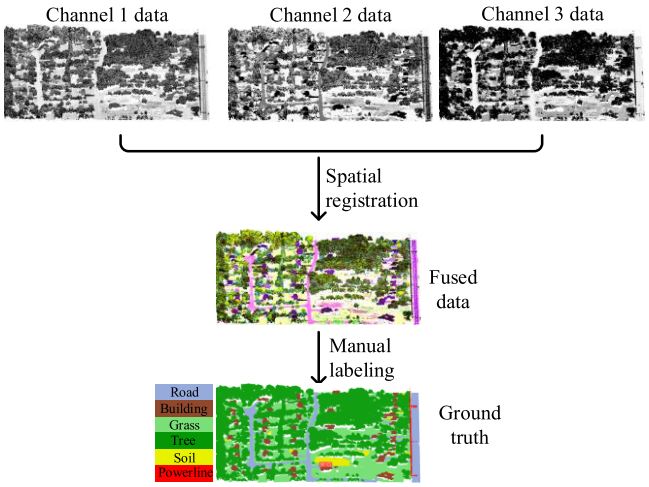


Fig. 6. Workflow of fusing and labeling data.

B. Comparative Methods

We evaluated the performance of the SCSQ-Net by comparing it with fully and weakly supervised deep learning networks for both MS-LiDAR data and conventional single-channel point clouds integrated with color information. We selected two fully supervised models, i.e., SE-PointNet++ [14] and FR-GCNet [15], designed specifically for MS-LiDAR point clouds, along with the SOTA networks, including RandLA-Net [13], DLA-Net [36], BAAF-Net [37], and BAF-LAC [38], for single-channel point clouds integrated with color information. Furthermore, we selected three weakly supervised models, i.e., FPWS-Net [22] for MS-LiDAR point clouds, as well as SQN [21] and PSD [23] for single-channel point clouds integrated with color information. The performance results of SE-PointNet++ and FR-GCNet were obtained from [15], as they used the same WS MS-LiDAR dataset. The MS-LiDAR point cloud classification results of the other models were obtained using their respective official codes, as well as the same training parameters and experimental environment as the SCSQ-Net.

C. Implementation Details

All experiments were conducted on a workstation with hardware consisting of an AMD EPYC 7302 CPU, 32 GB RAM, and 24 GB NVIDIA GeForce RTX 3090 GPU, under an Ubuntu 20.04 operating system. For model training, the batch size, initial learning rate, epoch, and decay rate per cycle were set to 2%, 0.001%, 100%, and 5%, respectively. The Adam optimizer was used. The number J of the kernel points in the SKPC module was set as 9, the number K_1 of the nearest neighbors of the input points was set as 20, the number K_2 of the query points of the labeled points was set as 9, and the number of the input points at a single batch was set as 65 536.

To further evaluate the performance of our SCSQ-Net, we used the following four evaluation metrics: $F1$-score to assess the accuracy performance of each class, and average $F1$-score, mean intersection over union (mIoU), and overall accuracy (OA) to evaluate the OA performance.

IV. Experimental Results

A. Results on the WS Dataset

Table I lists the quantitative results obtained by the comparative networks on the WS dataset. As listed in Table I, compared with the two fully supervised methods for MS-LiDAR point clouds, the SCSQ-Net demonstrated significant performance gains, with improvements of 8.22%, 14.93%, and 2.74% in $F1$-score, mIoU, and OA, respectively. Furthermore, all classes showed notable $F1$-score increases. These performance gains are because the SCSQ-Net used random sampling, rather than the FPS strategy used in the fully supervised SE-PointNet++ and FR-GCNet, to generate training samples. This random sampling provided a broader receptive field for the network, enhancing feature extraction and classification performance.

The SCSQ-Net also achieved comparable results to the SOTA methods, i.e., RandLA-Net, DLA-Net, BAAF-Net, and BAF-LAC, using only 0.1% labeled points. Notably, the SCSQ-Net obtained the best performance for the soil class with an $F1$-score of 46.27%. However, the classification accuracy for the powerline class was slightly lower due to its under-representation in the WS dataset.

In comparison with other weakly supervised networks, the SCSQ-Net outperformed SQN and PSD with $F1$-score gains of 11.06%–8.2%, and mIoU gains of 14.92%–9.49%, respectively. Significant improvements were observed for the road, soil, and powerline classes. This success is attributed to the SKPC module, which efficiently extracted local features, and the ADA module, which promoted comprehensive scene understanding. Conversely, the trilinear interpolation used in SQN failed to adequately extract the underlying supervised signals, and the dual-branch structure in PSD depended on the performance of the backbone.

When compared to FPWS-Net, the SCSQ-Net achieved modest improvements of 0.19%–0.62% in $F1$-score and mIoU, respectively. The most notable gains were for the building, tree, and grass classes. The pseudo-label strategy used in FPWS-Net generated much noise, limiting its ability to interpret distinct features effectively. In contrast, the SCSQ-Net leveraged precise supervised signals from sparsely labeled points, achieving superior performance on the WS dataset.

To better demonstrate the visualization results, we selected Area 11 from the three test scenes. Fig. 7 visually presents the ground truth and the results obtained by the comparative networks. As illustrated in Fig. 7, the point cloud classification results obtained by the SCSQ-Net were greatly consistent with the ground truth, except for some soil points. Visual inspection also indicated that the classification results of the SCSQ-Net contained few misclassifications, comparable to those of the fully supervised models. Compared with the weakly supervised networks, i.e., FPWS-Net, SQN, and PSD, the SCSQ-Net obtained great improvements for road and soil points. However, the SCSQ-Net had a slight performance degradation for class soil, compared with FPWS-Net.

To clearly analyze the classification performance of the SCSQ-Net, we selected the two areas marked by the black

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TABLE I
QUANTITATIVE RESULTS ON THE WS DATASET

Methods		F_1-score (%)						Average F_1-score (%)	mIoU (%)	OA (%)
		Road	Building	Tree	Grass	Soil	Powerline			
Fully supervised	SE-PointNet++ [14]	70.32	85.64	94.70	97.05	37.02	70.35	75.84	64.32	93.01
	FR-GCNet [15]	84.63	90.81	95.33	98.77	28.72	74.11	78.61	65.78	93.55
	RandLA-Net [13]	88.05	99.30	96.77	99.68	40.20	96.49	86.75	81.88	96.41
	DLA-Net [36]	80.95	99.34	93.72	99.80	18.90	98.93	81.94	77.01	94.23
	BAAF-Net [37]	88.29	99.45	96.66	99.75	42.76	95.13	87.00	81.90	96.39
	BAF-LAC [38]	88.02	99.46	96.52	99.79	42.80	98.92	87.59	83.17	96.30
Weakly supervised (0.1%)	SQN [21]	72.15	99.05	94.36	99.23	4.26	85.6	75.77	65.79	92.45
	PSD [23]	81.24	98.76	94.81	99.12	22.81	75.01	78.63	71.22	94.59
	FPWS-Net [22]	88.43	98.46	93.96	99.65	47.15	92.19	86.64	80.09	96.37
	SCSQ-Net	87.60	99.31	96.45	99.69	46.27	91.66	86.83	80.71	96.29

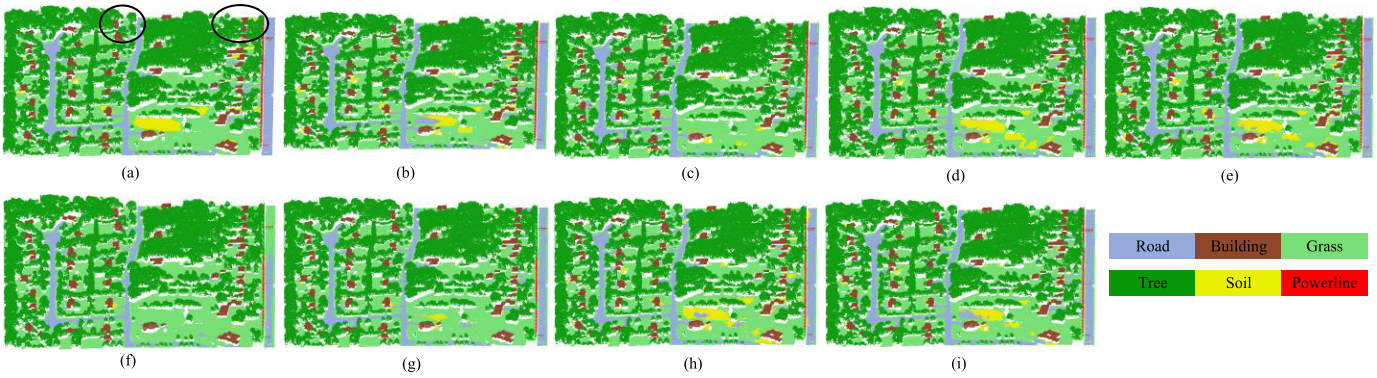


Fig. 7. Visual results of different fully supervised methods and weakly supervised methods on the WS dataset. (a) Ground truth. (b) RandLA-Net. (c) DLA-Net. (d) BAAF-Net. (e) BAF-LAC. (f) SQN. (g) PSD. (h) FPWS-Net. (i) SCSQ-Net.

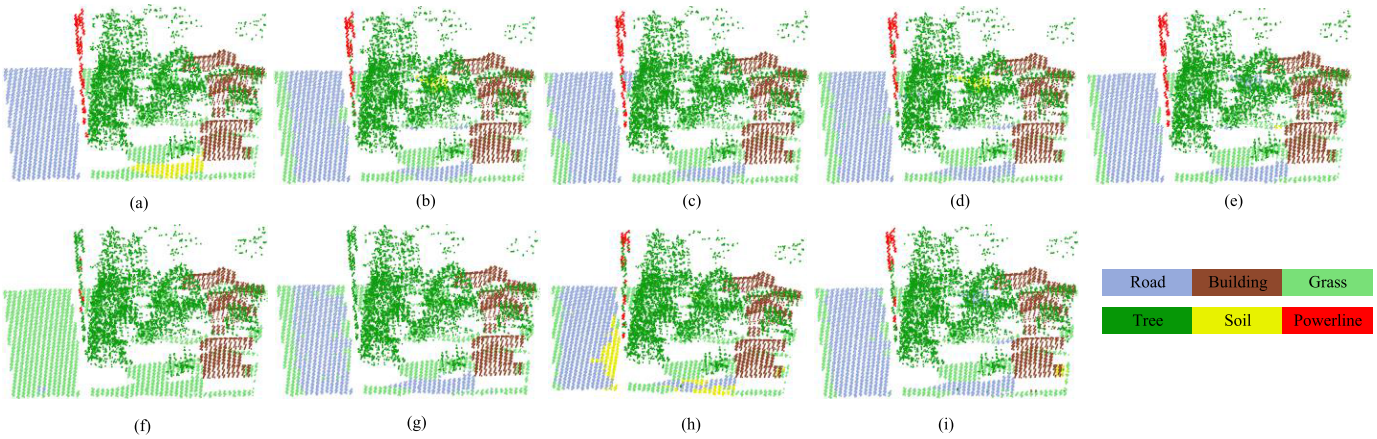


Fig. 8. Close-up visualization of the comparison methods. (a) Ground truth. (b) RandLA-Net. (c) DLA-Net. (d) BAAF Net. (e) BAF-LAC. (f) SQN. (g) PSD. (h) FPWS-Net. (i) SCSQ-Net.

ellipses from Area 11 [see Fig. 7(a)] for a close-up visual inspection. Figs. 8 and 9 display the classification results of the two selected areas, respectively. As shown in Fig. 8, except FPWS-Net, all networks failed to identify all soil points from the scene and misclassified the soil points as roads. The reason behind this phenomenon is that soil and road points lack distinct geometric features and exhibit the same-spectrum heterogeneity. Also, the SCSQ-Net and the other compared networks misclassified some road points at the edge of the scene as grass points. This occurred because the road points at the scene edge lacked sufficient neighborhood information, and their geometric features, e.g., unclear boundaries, were similar to those of class grass. Overall, the classification performance of the SCSQ-Net was roughly equivalent to those of the supervised methods. Compared to the weakly supervised networks, the SCSQ-Net accurately identified the majority of road, grass, and powerline points, while SQN also failed to correctly classify road and powerline points due to the sparse powerline points. Moreover, SQN relied solely on the supervised signals from labeled points without fully utilizing

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TABLE II
QUANTITATIVE RESULTS ON THE TOBERMORY DATASET

Methods		F_1-score (%)						Average F_1-score (%)	mIoU (%)	OA (%)
		Road	Building	Land	Tree	Water	Powerline			
Fully supervised	RandLA-Net [14]	87.57	94.88	95.99	99.42	99.02	92.82	94.95	89.54	98.03
	DLA-Net [36]	74.61	95.18	92.01	99.83	98.92	90.37	91.82	85.61	96.92
	BAAF-Net [37]	87.75	95.67	96.26	99.42	99.30	94.29	95.45	90.67	98.18
	BAF-LAC [38]	89.23	96.09	96.62	99.35	99.19	94.35	95.81	92.15	98.35
Weakly supervised (0.1%)	SQN [21]	67.34	93.42	91.52	98.85	97.67	78.51	87.88	80.31	96.29
	PSD [23]	70.01	88.54	93.19	99.09	97.73	88.61	89.53	82.63	96.72
	FPWS-Net [22]	83.26	89.70	95.07	98.91	97.18	84.49	91.44	85.35	97.32
	SCSQ-Net	83.98	91.15	93.13	99.54	99.06	83.88	91.79	85.46	96.94

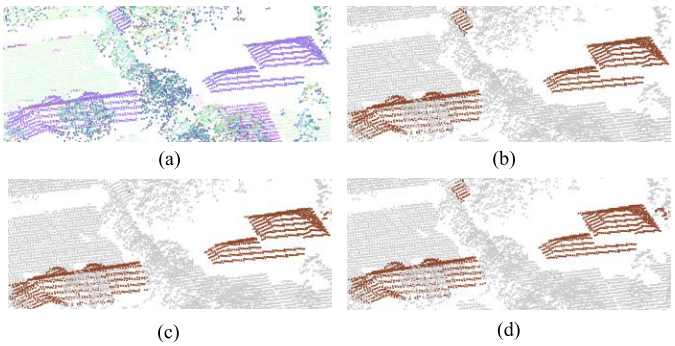


Fig. 9. Close-up visualization of buildings of the comparison methods. (a) MS-LiDAR. (b) Ground truth. (c) FPWS-Net. (d) SCSQ-Net.

unlabeled points. Although FPWS-Net correctly identified soil points from the scene, it misclassified some road points as soil due to noisy pseudo-labels generated during training.

To further demonstrate the superiority of the SCSQ-Net to FPWS-Net, we specifically selected several buildings for comparison (see Fig. 9). As shown in Fig. 9, in this subscene, buildings are obscured by tall and dense trees. In such a complex scenario, it is challenging to identify small buildings hidden under the trees. FPWS-Net failed to correctly classify the small buildings in this scene, whereas our SCSQ-Net correctly classified all buildings, demonstrating the capability and robustness of the SCSQ-Net in identifying small targets within complex point clouds.

B. Results on the Tobermory Dataset

To further validate the applicability and generalization of our method in the MS-LiDAR point cloud classification task, we conducted experiments on the Tobermory dataset, which has a higher point density than the WS dataset. The same network parameters used for the WS dataset were applied to the Tobermory dataset to ensure experimental consistency.

Table II lists the quantitative results of the comparative methods on the Tobermory dataset. Compared with the fully supervised methods, Our SCSQ-Net performed excellently on the overall evaluation metrics, even surpassing DLA-Net on certain classes such as road, land, and water. For the tree class, the SCSQ-Net achieved the second-highest $F1$-score among the fully supervised methods, falling behind DLA-Net by only 0.29%. It was observed that the Tobermory dataset contains a rich variety of road, land, tree, and water points. Therefore, even with only 0.1% of labeled points, the SCSQ-Net is able to achieve highly accurate point cloud classification results.

Compared with the weakly supervised methods, the SCSQ-Net achieved the highest accuracies on the road, tree, and water classes. This indicates that the SCSQ-Net is better able to integrate the supervised signals from the neighborhood of labeled points. In comparison with FPWS-Net, the SCSQ-Net showed $F1$-score improvements of 0.72%, 1.45%, 0.63%, and 1.88% on the road, building, tree, and water classes, respectively. Overall, the SCSQ-Net improved the metrics by 0.35% on average $F1$-score and 0.11% on mIoU compared with FPWS-Net. The SCSQ-Net primarily used only labeled points and their neighborhoods, avoiding the extensive reliance on unlabeled point information used in FPWS-Net. The comparative results demonstrated that our SCSQ-Net is a promising solution to the MS-LiDAR point cloud classification task using only 0.1% labeled points.

TABLE III
ABLATION STUDY RESULTS

Model	EF	SKPC	ADA	Affinity Loss	Average F_1-score (%)	mIoU (%)
A		√	√	√	85.78	79.33
B	√		√	√	85.29	78.90
C	√	√		√	85.99	80.04
D	√	√	√		83.86	77.98
SCSQ-Net	√	√	√	√	86.83	80.71

C. Ablation Study

In this section, to analyze the key components of the SCSQ-Net, we decomposed it into several models. All experiments were conducted on the WS dataset. The results are shown in Table III.

1) Effect of the EF Module: We removed the EF module from the SCSQ-Net, and the resultant model was named model A. The performance of model A decreased by 1.05% on average $F1$-score and 1.38% on mIoU, compared with the

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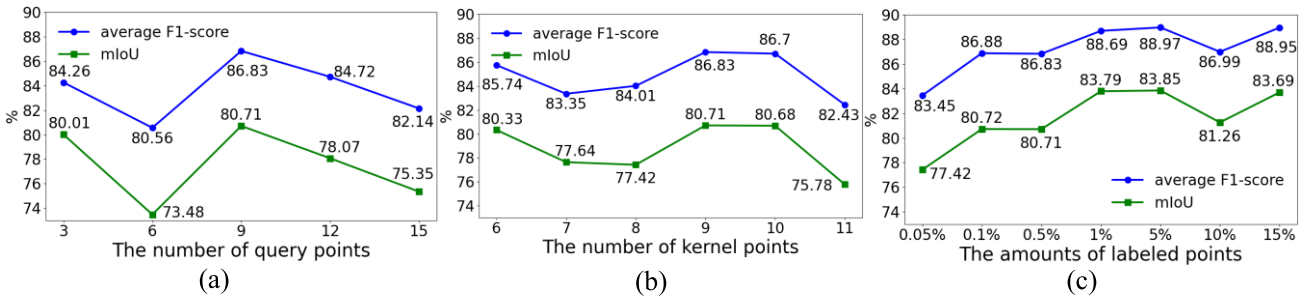


Fig. 10. Sensitivity analysis of the SCSQ-Net. (a) Classification accuracies using different numbers of query points. (b) Classification accuracies using different numbers of kernel points. (c) Classification accuracies using different amounts of labeled points.

SCSQ-Net. The EF module significantly improved the model performance, because it fully considered the geometric and feature distance relationships between the central point and its neighboring points in the SKPC module. This demonstrated that the SKPC module provided rich features, thereby offering the network precise classification signals.

2) Effect of the SKPC Module: In the encoding path and the semantic inference path of the SCSQ-Net, we used two independent kernel point convolutions without sharing kernel point positions and weight matrices, and the resultant model was named model B. Comparing model B with the SCSQ-Net, the performance decreased by 1.54% on average $F1$-score and 1.81% on mIoU. This indicated that, during the semantic inference path, SCSQ-Net using the kernel point positions and weight matrices from the encoding stage was capable of accurately aggregating the neighborhood information of labeled points, providing more beneficial information to the weakly supervised network.

3) Effect of the ADA Module: We removed the ADA module from the SCSQ-Net, and the resultant model was named model C. Compared with the SCSQ-Net, the accuracy obtained by model C decreased by 0.84% on average $F1$-score and 0.67% on mIoU. This demonstrated that the ADA module effectively prevented the SCSQ-Net from focusing too much on some easily distinguishable regions with obvious features. The ADA module activated the entire scene, enhancing its ability to recognize complex targets.

4) Effect of the Affinity Loss: We removed the affinity loss from the SCSQ-Net, training the model only with weight cross-entropy loss and Lovasz–Softmax loss, the resultant model was named model D. Note that the point cloud classification performance of model D decreased by 3.36% on average $F1$-score and 2.73% on mIoU. The affinity loss established feature affinities between labeled points, which reduced intra-class distance and increased interclass distance, and promoted the propagation of supervised signals from labeled to unlabeled points. The affinity loss fully utilized the labeled points, thereby improving the performance of our SCSQ-Ne on the MS-LiDAR point clouds.

D. Sensitivity Analysis of Hyperparameters

To investigate the sensitivity of the SCSQ-Net to key hyperparameters, i.e., the number of query points, the number of kernel points, and the number of labeled points, we conducted sensitivity analysis experiments on the WS dataset.

First, to determine the optimal number of query points, we set the number of kernel points in the SKPC module as 9 and varied the number of query points from 3 to 15 at an interval of 3. As seen in Fig. 10(a), the classification performance changed significantly with the increase in the number of query points. Observed that the lowest performance occurred with an average $F1$-score of 80.56% and a mIoU of 73.48% when the number of query points was set to 6, while the best results were achieved with an average $F1$-score of 86.83% and a mIoU of 80.71% when the number of query points was set to 9. To trade off the aggregation of the sufficient neighboring features and the interference of different categories, we set the number of query points to 9. It is suggested that the optimal number of query points may vary across datasets based on point cloud density and class distribution.

Next, we set the number of query points to 9 and varied the number of kernel points in the SKPC module from 6 to 11 at an interval of 1. As shown in Fig. 10(b), in terms of mIoU and average $F1$-score, the MS-LiDAR classification results obtained by the SCSQ-Net fluctuated with the change of the number of kernel points, decreasing from 6 to 8, increasing from 8 to 9, and then decreasing again from 9 to 11. The performance changes are mainly due to the reliance of both the encoding path and the semantic inference path on the performance of the SKPC module.

Lastly, we assessed the impact of labeled points on the MS-LiDAR classification performance by varying the labeling rate from 0.05% to 15%, when the numbers of query and kernel points were both set to 9. As seen in Fig. 10(c), the performance was improved slightly with an increase in the labeling rate, but the performance then decreased when the labeling rate was larger than 10%. The performance degradation might be because the more the labeled points are used, the more features from different classes the SKPC module aggregates, thus generating more noise. To balance performance and labeling efforts, we used a labeling rate of 0.1% for model training.

V. Conclusion

In this article, we proposed a novel weakly supervised network, named SCSQ-Net, specifically designed for MS-LiDAR point cloud classification tasks. In our SCSQ-Net, we first proposed an SKPC module, which shared the kernel point positions and weights for both the encoding path and the

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semantic inference path. The SKPC module fully extracted high-dimensional features from the MS-LiDAR point clouds while better leveraging semantic similarity to integrate neighborhood information from labeled points, providing the network with precise and rich supervised signals. Next, we introduced an ADA block, which ensured the model activated across the entire scene, avoiding overemphasis on certain regions due to sparse supervised signals. Finally, to fully utilize the sparse labeled points and comprehensively supervise network training, we designed a composite loss function by integrating the point affinity loss, the weighted cross-entropy loss, and the Lovasz–Softmax loss together.

Using only 0.1% labeled points, the SCSQ-Net achieved an average $F1$-score of 86.83%, a mIoU of 80.71%, and an OA of 96.29% on two MS-LiDAR datasets (WS and Tobermory). These results demonstrate the effectiveness of the SCSQ-Net in significantly reducing the manual labeling workload for large-scale MS-LiDAR datasets while maintaining high classification performance comparable to fully supervised models. Despite its strong performance, the SCSQ-Net requires hyperparameter adjustments tailored to the density and characteristics of different point clouds. In the future, we plan to design more efficient methods to achieve automatic adaptation to different densities and labeling rates in weakly supervised networks, aiming to achieve even better classification results.

REFERENCES

[1] P. Hartzell, C. Glennie, K. Biber, and S. Khan, "Application of multispectral LiDAR to automated virtual outcrop geology," *ISPRS J. Photogramm. Remote Sens.*, vol. 88, pp. 147–155, Feb. 2014.

[2] A. Collin, B. Long, and P. Archambault, "Merging land-marine realms: Spatial patterns of seamless coastal habitats using a multispectral LiDAR," *Remote Sens. Environ.*, vol. 123, pp. 390–399, Aug. 2012.

[3] A. Shaker, W. Y. Yan, and P. E. LaRocque, "Automatic land-water classification using multispectral airborne LiDAR data for near-shore and river environments," *ISPRS J. Photogramm. Remote Sens.*, vol. 152, pp. 94–108, Jun. 2019.

[4] M. Dalponte, L. T. Ene, T. Gobakken, E. Næsset, and D. Gianelle, "Prediction of forest attributes with multispectral LiDAR data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Valencia, Spain, Jul. 2018, pp. 7528–7531.

[5] W. Y. Yan, K. van Ewijk, P. Treitz, and A. Shaker, "Effects of radiometric correction on cover type and spatial resolution for modeling plot level forest attributes using multispectral airborne LiDAR data," *ISPRS J. Photogramm. Remote Sens.*, vol. 169, pp. 152–165, Nov. 2020.

[6] N. Ekhtari, C. Glennie, and J. C. Fernandez-Diaz, "Classification of airborne multispectral LiDAR point clouds for land cover mapping," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 6, pp. 2068–2078, Jun. 2018.

[7] S. Pan et al., "Land-cover classification of multispectral LiDAR data using CNN with optimized hyper-parameters," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 241–254, Aug. 2020.

[8] Y. Yu et al., "Land cover classification of multispectral LiDAR data with an efficient self-attention capsule network," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[9] R. Q. Charles, H. Su, M. Kaichun, and L. J. Guibas, "PointNet: Deep learning on point sets for 3D classification and segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 77–85.

[10] C. R. Qi, L. Yi, H. Su, and L. J. Guibas, "PointNet++: Deep hierarchical feature learning on point sets in a metric space," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst.*, Dec. 2017, pp. 5105–5114.

[11] Y. Wang, Y. Sun, Z. Liu, S. E. Sarma, M. M. Bronstein, and J. M. Solomon, "Dynamic graph CNN for learning on point clouds," *ACM Trans. Graph.*, vol. 38, no. 5, pp. 1–12, 2019.

[12] H. Thomas, C. R. Qi, J.-E. Deschaud, B. Marcotegui, F. Goulette, and L. Guibas, "KPConv: Flexible and deformable convolution for point clouds," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6410–6419.

[13] Q. Hu et al., "RandLA-Net: Efficient semantic segmentation of large-scale point clouds," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11105–11114.

[14] Z. Jing et al., "Multispectral LiDAR point cloud classification using SE-PointNet++," *Remote Sens.*, vol. 13, no. 13, p. 2516, Jun. 2021.

[15] P. Zhao et al., "Airborne multispectral LiDAR point cloud classification with a feature reasoning-based graph convolution network," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 105, Dec. 2021, Art. no. 102634.

[16] D. Li et al., "Building extraction from airborne multi-spectral LiDAR point clouds based on graph geometric moments convolutional neural networks," *Remote Sens.*, vol. 12, no. 19, p. 3186, Sep. 2020.

[17] D. Li et al., "AGFP-Net: Attentive geometric feature pyramid network for land cover classification using airborne multispectral LiDAR data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 108, Apr. 2022, Art. no. 102723.

[18] K. Xiao, J. Qian, T. Li, and Y. Peng, "Multispectral LiDAR point cloud segmentation for land cover leveraging semantic fusion in deep learning network," *Remote Sens.*, vol. 15, no. 1, p. 243, Dec. 2022.

[19] J. Yang, B. Luo, R. Gan, A. Wang, S. Shi, and L. Du, "Multiscale adjacency matrix CNN: Learning on multispectral LiDAR point cloud via multiscale local graph convolution," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 855–870, 2024.

[20] X. Xu and G. H. Lee, "Weakly supervised semantic point cloud segmentation: Towards 10× fewer labels," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 13703–13712.

[21] Q. Hu et al., "SQN: Weakly-supervised semantic segmentation of large-scale 3D point clouds," 2021, *arXiv:2104.04891*.

[22] K. Chen et al., "A feature perturbation weakly supervised learning network for airborne multispectral LiDAR pointcloud classification," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 127, Mar. 2024, Art. no. 103683.

[23] Y. Zhang, Y. Qu, Y. Xie, Z. Li, S. Zheng, and C. Li, "Perturbed self-distillation: Weakly supervised large-scale point cloud semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 15500–15508.

[24] J. Wei, G. Lin, K.-H. Yap, F. Liu, and T.-Y. Hung, "Dense supervision propagation for weakly supervised semantic segmentation on 3D point clouds," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 6, pp. 4367–4377, Jun. 2024.

[25] L. Jiang et al., "Guided point contrastive learning for semi-supervised point cloud semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 6403–6412.

[26] M. Li et al., "HybridCR: Weakly-supervised 3D point cloud semantic segmentation via hybrid contrastive regularization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 14910–14919.

[27] Y. Lin, G. Vosselman, and M. Y. Yang, "Weakly supervised semantic segmentation of airborne laser scanning point clouds," *ISPRS J. Photogramm. Remote Sens.*, vol. 187, pp. 79–100, May 2022.

[28] J. Wei, G. Lin, K.-H. Yap, T.-Y. Hung, and L. Xie, "Multi-path region mining for weakly supervised 3D semantic segmentation on point clouds," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4383–4392.

[29] P. Wang and W. Yao, "A new weakly supervised approach for ALS point cloud semantic segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 188, pp. 237–254, Jun. 2022.

[30] X. Lei et al., "WSPointNet: A multi-branch weakly supervised learning network for semantic segmentation of large-scale mobile laser scanning point clouds," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 115, Dec. 2022, Art. no. 103129.

[31] P. Wang, W. Yao, and J. Shao, "One class one click: Quasi scene-level weakly supervised point cloud semantic segmentation with active learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 204, pp. 89–104, Oct. 2023.

[32] A. Deng, Y. Wu, P. Zhang, Z. Lu, W. Li, and Z. Su, "A weakly supervised framework for real-world point cloud classification," *Comput. Graph.*, vol. 102, pp. 78–88, Feb. 2022.

[33] O. Unal, D. Dai, and L. Van Gool, "Scribble-supervised LiDAR semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 2687–2697.

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[34] M. Berman, A. R. Triki, and M. B. Blaschko, "The Lovász-Softmax loss: A tractable surrogate for the optimization of the intersection-over-union measure in neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4413–4421.

[35] V. Wichmann, M. Bremer, J. Lindenberger, M. Rutzinger, C. Georges, and F. Petrini-Monteferri, "Evaluating the potential of multispectral airborne LiDAR for topographic mapping and land cover classification," *ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 2, pp. 113–119, Aug. 2015.

[36] Y. Su et al., "DLA-Net: Learning dual local attention features for semantic segmentation of large-scale building facade point clouds," *Pattern Recognit.*, vol. 123, Mar. 2022, Art. no. 108372.

[37] S. Qiu, S. Anwar, and N. Barnes, "Semantic segmentation for real point cloud scenes via bilateral augmentation and adaptive fusion," 2021, *arXiv:2103.07074*.

[38] H. Shuai, X. Xu, and Q. Liu, "Backward attentive fusing network with local aggregation classifier for 3D point cloud semantic segmentation," *IEEE Trans. Image Process.*, vol. 30, pp. 4973–4984, 2021.



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